N4000-29 - Dielectric Properties Table

LAMINATE:

Thickness (inch)	&	Tol.	Construction					RC%	Nominal Thickness (inch)	Nominal Thickness (µm)	2 GHz Dk	2 GHz Df	10 GHz Dk	10 GHz Df
0.0020	±	0.0005	1	106				71%	0.0021	52	3.70	0.0201	3.62	0.0216
0.0025	±	0.0005	1	1080				59%	0.0026	66	4.00	0.0185	3.92	0.0199
0.0030	±	0.0005	1	1080				64%	0.0031	78	3.86	0.0193	3.79	0.0206
0.0030	±	0.0005	1	2113				46%	0.0031	78	4.39	0.0164	4.32	0.0176
0.0035	±	0.0005	2	106				63%	0.0031	79	3.88	0.0191	3.81	0.0205
0.0035	±	0.0005	1	2113				52%	0.0036	92	4.19	0.0175	4.12	0.0188
0.0040	±	0.0005	1	2116				46%	0.0041	103	4.39	0.0164	4.33	0.0176
0.0040	±	0.0005	1	106	+	1	1080	61%	0.0043	109	3.94	0.0189	3.86	0.0202
0.0045	±	0.0005	1	2116				51%	0.0046	117	4.23	0.0172	4.16	0.0185
0.0050	±	0.0007	1	2116				55%	0.0051	130	4.11	0.0179	4.04	0.0192
0.0050	±	0.0007	1	106	+	1	2113	55%	0.0051	131	4.11	0.0179	4.03	0.0192
0.0050	±	0.0007	2	1080				59%	0.0052	133	4.00	0.0185	3.92	0.0199
0.0060	±	0.0007	2	1080				64%	0.0061	156	3.86	0.0193	3.79	0.0206
0.0060	±	0.0007	2	2113				46%	0.0062	156	4.39	0.0164	4.32	0.0176
0.0060	±	0.0007	1	1080	+	1	2113	55%	0.0062	158	4.11	0.0179	4.04	0.0192
0.0070	±	0.0010	2	2113				52%	0.0072	183	4.19	0.0175	4.12	0.0188
0.0070	±	0.0010	1	1080	+	1	2116	54%	0.0073	186	4.13	0.0178	4.06	0.0191
0.0080	±	0.0010	2	2116				46%	0.0081	207	4.39	0.0164	4.33	0.0176
0.0080	±	0.0010	1	2113	+	1	2116	51%	0.0082	208	4.23	0.0172	4.16	0.0185

N4000-29 - Dielectric Properties Table

LAMINATE (Continued):

Thickness (inch)	&	Tol.	Construction					RC%	Nominal Thickness (inch)	Nominal Thickness (µm)	2 GHz Dk	2 GHz Df	10 GHz Dk	10 GHz Df
0.0090	±	0.0010	2	2116				51%	0.0092	234	4.23	0.0172	4.16	0.0185
0.0100	±	0.0010	2	2116				55%	0.0103	261	4.11	0.0179	4.04	0.0192
0.0120	±	0.0015	2	1080	+	1	7628	49%	0.0120	305	4.28	0.0169	4.22	0.0182
0.0140	±	0.0015	2	2113	+	1	7628	48%	0.0148	376	4.33	0.0167	4.26	0.0180
0.0180	±	0.0015	2	1080	+	2	7628	44%	0.0188	478	4.45	0.0160	4.39	0.0172
0.0200	±	0.0025	4	2116				55%	0.0205	521	4.11	0.0179	4.04	0.0192
0.0240	±	0.0025	2	2116	+	2	7628	48%	0.0254	646	4.32	0.0168	4.25	0.0180
0.0250	±	0.0025	2	1080	+	3	7628	42%	0.0256	651	4.50	0.0157	4.44	0.0169
0.0280	±	0.0025	2	2116	+	3	7628	41%	0.0286	726	4.57	0.0154	4.51	0.0166
0.0310	±	0.0040	2	1080	+	4	7628	42%	0.0325	824	4.53	0.0156	4.47	0.0168
0.0380	±	0.0040	2	1080	+	5	7628	40%	0.0385	977	4.59	0.0153	4.53	0.0165
0.0590	±	0.0050	2	1080	+	8	7628	40%	0.0597	1516	4.58	0.0153	4.52	0.0165
0.0620	±	0.0050	2	2113	+	8	7628	42%	0.0640	1626	4.53	0.0156	4.47	0.0168

N4000-29 - Dielectric Properties Table

PREPREG:

Glass	RC%	Thickness (inches)	Thickness (µm)	2 GHz Dk	2 GHz Df	10 GHz Dk	10 GHz Df
106	75%	0.0025	62	3.61	0.0206	3.53	0.0221
1080	65%	0.0032	81	3.84	0.0194	3.76	0.0208
1080	71%	0.0040	101	3.70	0.0202	3.62	0.0216
2113	57%	0.0041	105	4.04	0.0183	3.97	0.0196
2113	61%	0.0047	118	3.94	0.0189	3.86	0.0202
2116	56%	0.0054	136	4.07	0.0181	4.00	0.0194
2116	59%	0.0058	148	3.99	0.0186	3.91	0.0199
7628	44%	0.0077	195	4.45	0.0160	4.38	0.0173

These laminate and prepreg items are representative of available materials. These items and their associated values may not be regularly available and additional lead times or minimum order quantities could apply. Please contact our office or your local technical representative for assistance with choosing the best material for your design and application.